

Challenges and Initial Ideas for Scaled Natural Jumping in Large Immersive Environments

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ABSTRACT

Navigating expansive immersive virtual environments remains a fundamental challenge in Virtual Reality, necessitating locomotion metaphors that balance efficiency with physical presence. Natural jumping has emerged as a promising solution to bridge the gap between the intuitiveness of natural walking and the efficiency of teleportation. However, recent studies have shown that significant challenges remain in physical fatigue, simulator sickness, and spatial awareness. In this position paper, we analyze the current state of natural jumping research, identify critical design limitations, and propose a multi-stage research agenda. By conducting systematic reviews, controlled user studies, and functional case studies, we aim to contribute evidence-based guidelines that establish scaled natural jumping as a safe, effective, and intuitive locomotion hybrid for the future of VR navigation.

Index Terms: Virtual Reality, Locomotion, Jumping, Natural Jumping

1 INTRODUCTION

Jumping is an instinctive human movement, central to activities ranging from athletic performance to everyday navigation. In digital entertainment, video games frequently leverage jumping as a core mechanic, sometimes mimicking natural biomechanics and other times granting superhuman capabilities, such as multi-jumping or leaping across vast distances [1, 2, 5, 20]. Consequently, recent research has investigated natural jumping as a locomotion metaphor in Virtual Reality (VR), yielding promising results in sports training, immersive gaming, and general spatial navigation [6, 9, 10, 26].

While natural walking remains the most immersive locomotion method by mapping physical steps directly to the virtual world, it is fundamentally constrained by the limitations of physical tracking space [19]. We propose that natural jumping can serve as a seamless extension of natural walking, enabling users to traverse large-scale immersive virtual environments (IVEs) without sacrificing immersion or spatial awareness. Integrating natural walking for local movement with “scaled jumping” for long-distance traversal could provide a highly efficient and effective locomotion hybrid [27]. Despite existing studies on verticality in VR, a significant research gap remains in the usability, comfort, and precision of scaled natural jumping as a navigation tool for expansive IVEs [9, 14, 26].

In this position paper, we examine the viability of scaled natural jumping as a navigation metaphor. The following sections provide an overview of current VR locomotion strategies (Section 2), a detailed analysis of the jumping metaphor (Section 3) and its inherent limitations and design challenges (Section 4), and an outline of the research questions and initial ideas we aim to address in our future research (Section 5).

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2 LOCOMOTION

Locomotion is a fundamental interaction in VR, allowing users to traverse IVEs by transforming their virtual viewpoint [3, 18, 24]. Through various locomotion techniques, users can explore virtual spaces in ways that are dictated by the specific navigation method, the characteristics of the environment, and the primary objectives of the application [18].

Natural walking is considered the most natural locomotion metaphor, producing a significantly higher sense of presence compared to other techniques [19, 22, 23]. However, the use of physical walking is inherently restricted by the size of the available tracking place [19]. While reorientation techniques—such as redirected walking, motion compression, and scaled translational gains attempt to overcome these spatial limitations, they often remain inefficient when applied to expansive virtual environments [8, 17, 15, 19, 25].

To address the need for efficiency, alternative solutions like teleportation, gaze-directed locomotion, and gesture-based movement have been developed. Teleportation, in particular, has emerged as a dominant approach in commercial and gaming applications due to its superior usability [4]. Despite their efficiency, these methods often detract from the user’s sense of presence [7, 21]. The “unnaturalness” of being instantaneously relocated breaks the continuity of the experience and can diminish the feeling of being physically situated within the world [21]. In order to combine the high presence of physical movement with the efficiency of artificial navigation, natural jumping has been introduced as a promising alternative [9].

3 NATURAL JUMPING

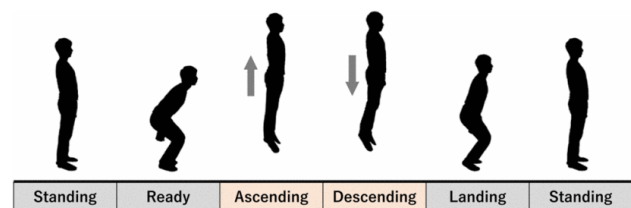


Figure 1: The five states of jumping. **Left to right:** Standing, Ready, Ascending, Descending, Landing, and back to Standing [9]. A jump detection algorithm is used to determine the start and end of a physical jump.

Natural jumping, similar to natural walking, could be defined as mapping a user’s physical jumping action in the real world to a corresponding movement within an immersive environment [19]. Hayashi et al. originally proposed the concept of Redirected Jumping (RDJ), investigating detection thresholds for the imperceptible manipulation of these movements [9]. In this study, a natural jump was observed to be a series of physical states: standing, ready, ascending, descending, landing, and returning to the standing position as can be seen in Figure 1. To identify these states, the authors established tracking guidelines using the user’s head, waist, and feet

[9]. This framework has since been adopted by various researchers to facilitate jump detection in virtual environments [11, 13, 26]. Furthermore, Hayashi et al. demonstrated that natural jumps can be manipulated across three axes: distance, height, and rotation. Building on this, Li et al. explored the interaction between distance and height gains to determine the limits of their imperceptibility when applied simultaneously [13].

To implement these manipulations, many recent studies employ Kim et al.'s method to apply visual gains to a user's jump [12]. Although Kim et al.'s study utilized a specialized cable-driven gantry system to simulate reduced gravity, their method of scaling the vertical trajectory has become foundational for natural jump manipulation. Subsequent research has continued to employ these gain-based techniques to modify natural jumping mechanics even within standard VR setups, using them to enhance vertical reach or adjust the perceived intensity of the movement [9, 11, 16, 26].

3.1 Use Cases of Natural Jumping

Building on these principles of natural jumping, several studies have explored how jumping mechanics can be leveraged to enhance the user experience. These works not only demonstrate the practical application of natural jumping but also propose critical design guidelines while highlighting the physical and technological limitations inherent in natural jumping.

3.1.1 Exergames

Due to its importance as a core motor skill, several studies have investigated the integration of jumping into VR exergames to enhance strength training and rehabilitation.

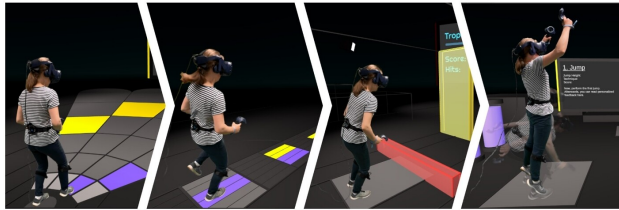


Figure 2: Depiction of the four different levels of *JumpExTra*, an exergame developed to motivate lower body exercise [6]. Players perform various movements in the four levels. **Left to right:** taps, hops, jumps, and maximal vertical jumps.

Cmentowski et al. developed *JumpExTra VR*, an exergame designed to motivate lower-body exercise [6]. The application consists of four sequential levels: tapping on colored blocks, in-place hopping, dodging obstacles, and performing maximal vertical jumps as shown in Figure 2. In an exploratory study with 25 participants, the authors found that while the exergame was considered enjoyable and a “worthy challenge” by domain experts, it also led to increased simulator sickness, specifically nausea, and physical fatigue. Participants reported that these factors, along with the strenuous nature of the exercises, negatively impacted their motivation for long-term play. Consequently, the authors proposed five design implications for lower-body VR activities: mitigating “unintended forward drift,” addressing technological tracking constraints, managing safety concerns, incorporating rhythmic elements for fluent movement patterns, and adapting difficulty to the player’s fitness level.

In contrast, Ioannou et al. investigated the effects of scaling physical movements within an immersive “jump and run” exergame [10]. Participants performed vertical jumps and ran in place, while the system augmented their forward and vertical movement by up to 2x. This study indicated that augmenting physical abilities through scaling made participants more motivated to exercise compared to

non-augmented conditions, as they perceived their enhanced physical capabilities as a source of increased engagement.

3.1.2 Manipulated Navigation

Recent research has investigated the use of amplified and manipulated natural jumping as a means to enhance vertical navigation in IVEs as well.

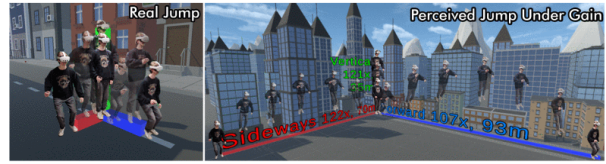


Figure 3: Scaled natural jumping with amplified gains [26]. **Left:** The average distance and height physically jumped by participants. **Right:** The perceived distance and height after applying maximal gain.

Kim et al. explored the concept of scaled jumping within a cable-driven system designed to simulate reduced gravity environments, such as those on the Moon or Mars [12]. By offloading a portion of a user’s weight through a gantry system, the authors applied visual gains to vertical movement to determine the range in which virtual jumps can be scaled without users detecting a discrepancy. Through a two-alternative forced-choice experiment, they identified a tolerable scaling range and subsequently developed a “retargeted jump” application, allowing users to jump onto virtual objects on a flat physical floor by manipulating gains specifically during the ascending and descending phases. Their findings indicate that while users could successfully land on virtual platforms using these gains, the descending phase was perceived as more challenging and significant for maintaining a sense of presence. Furthermore, they observed that users naturally increased their physical jumping effort when attempting to land on higher virtual objects.

Expanding on these manipulation techniques, Ogawa et al. developed *PseudoJumpOn* to simulate jumping onto virtual steps using standard VR hardware [16]. The system employs gain manipulation, applying higher gains during ascent than descent, and peak shifting, which delays the virtual peak height of the jump to mimic the trajectory of a step-up jump. In a study with 20 participants, the authors found that most conditions received positive scores for reality and naturalness, noting that participants were generally tolerant of high gains even when jumping onto unreachable virtual heights. Based on these findings, the authors recommended presenting a clear view of reaching the platform to enhance reality and utilizing moderate gains to lower perceived unnaturalness. They also cautioned against repetitive jumping due to user fatigue and observed that female participants reported lower reality scores, likely due to differences in physical jumping capacity.

More recently, Wilson et al. [26] investigated the effects of “superhuman” translational gains applied to forward, vertical, and sideways jumping as shown in Figure 3. Using a staircase approach to identify maximum comfortable thresholds, the authors found that most participants remained comfortable with gains exceeding 100x, reporting a sense of physical enhancement while maintaining a strong sense of jump ownership. The results suggest that such extreme scaling is effective and usable, potentially serving as an efficient extension to other locomotion metaphors for traversing large virtual distances. However, the study also identified several limitations and safety concerns, particularly regarding “forward drift” and the fear of real-world collisions when performing high-gain dynamic movements. Based on these findings, the authors presented several design guidelines: high gains (>100x) can be applied freely if preceded by an initial calibration, but more

modest gains (20x–40x) are recommended for vertical and sideways jumps to better retain jump ownership. Additionally, they noted that safety should be addressed through expanded boundary feedback or spatial awareness tools that provide pre-emptive warnings for ballistic movements.

Collectively, these studies demonstrate that while manipulated jumping is an effective tool for both navigation and exercise, its successful implementation depends on a careful balance between augmentation and physical safety. The recurring design guidelines across these works serve to mitigate the inherent risks of simulator sickness and physical fatigue. Understanding these limitations is critical for developing jump-centered environments that remain safe and engaging over extended periods of use.

4 LIMITATIONS AND DESIGN CHALLENGES

As discussed in Section 3, natural jumping presents several promising use cases for immersive environments. However, to solidify its integration into VR, several limitations and design challenges must be addressed.

First, jumping can cause significant physical fatigue, a phenomenon observed in both exergames and navigation studies requiring repetitive jumping [6, 9, 10, 11]. While fatigue is a desired outcome in exercise-based applications, it is a significant barrier for general navigation where immersion and long-term usability are prioritized.

Motion sickness and general discomfort also remain prevalent challenges [6, 16]. Although Wilson et al. reported a lack of simulator sickness in their high-gain study, this may be attributed to learning effects resulting from their staircase evaluation approach [26]. This suggests that as users gain experience with jumping metaphors, they may become accustomed to the sensation, leading to improved comfort and reduced sickness over time.

Finally, safety concerns remain a primary challenge for jumping in VR [6, 26]. Prior studies have proposed guidelines such as expanded boundary feedback and advanced spatial awareness tools [26]. However, the nature of physical jumping is inherently more dangerous than natural walking; once a user initiates a jump, they continue to travel through the air until they land, a process performed while their vision of the real world is occluded by an HMD. While a spacious, obstacle-free environment is a necessary baseline, we suggest that further research into pre-travel information is required. This could include warning systems for potential real-world obstacles or cross-reality techniques that allow users to “peek” into the physical environment before initiating a jump.

5 RESEARCH QUESTIONS AND INITIAL IDEAS TO INVESTIGATE

In order to address the limitations and design challenges mentioned in the previous section (Section 4), we consider the following research questions as the objective of our research:

- **RQ1: How can we enhance navigation of complex and large-scale IVEs using natural jumping?**

This question explores the balance between physical exertion, user experience, and virtual travel, specifically investigating how natural jumping metaphors can allow users to navigate large IVEs effectively, efficiently, and comfortably.

- **RQ2: How can we mitigate motion sickness while utilizing natural jumping?**

We aim to identify the specific visual and temporal factors during the jump trajectory that trigger motion sickness and explore solutions to minimize it in order to promote a more comfortable experience during natural jumping.

- **RQ3: How can we avoid disorientation and maintain spatial awareness while using natural jumping to navigate large complex IVEs?**

This objective focuses on maintaining the user’s cognitive map and sense of orientation when jumping across large virtual distances or between occluded viewpoints.

To effectively answer our research questions, we will focus on a small number of specific research topics.

5.1 Systematic review on natural jumping

A comprehensive survey of existing natural jumping implementations will allow us to systematically categorize current limitations and design challenges. This review will identify critical gaps in the literature and provide a structured knowledge base to benefit the broader VR research community.

5.2 User Studies evaluating Natural Jumping Metaphors

To better support navigating the complex and large-scale IVEs, usability studies will be conducted to find the balance between navigation efficiency and usability, including fatigue and comfort.

5.3 Case Studies of Natural Jumping Techniques

We intend to deploy optimized jumping techniques within functional case studies to evaluate their utility beyond abstract testing. These investigations will focus on real-world applications, such as architectural walkthroughs or complex environment traversal, to observe how the metaphors function in practical use.

5.4 Establishing Design Guidelines for Natural Jumping

The findings synthesized from the systematic review, user studies, and case studies will be used to establish a comprehensive set of design guidelines. These evidence-based recommendations will assist the VR community in developing safe, effective, and intuitive jumping-based navigation systems.

6 CONCLUSION

In this position paper, we examined the potential of scaled natural jumping as a locomotion metaphor for navigating large IVEs. By analyzing current use cases and identifying critical limitations such as physical fatigue, simulator sickness, and safety concerns, we highlight the need for research that moves beyond simple movement amplification to focus on user comfort and spatial orientation. To address these gaps, we propose a research agenda comprising a systematic review, controlled user studies, and functional case studies. This future work aims to establish evidence-based design guidelines that will allow the VR community to implement natural jumping as a safe, intuitive, and superhuman navigation technique.

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